2026–2027 AIDaS Seed Grant Program

Self-Repairing AI Agents on the Wayne State Grid: From Labeled CT to C-Arm Level Localization, with Medical Trainees Driving the Agents

Strategic areas	Foundational AI (area 1, primary; Aim 1): the reliability of agentic systems, studied in an agent held to pass/fail tests fixed before each run and to the cluster's safety rules. AI for Education and Workforce Training (area 3; Aim 2): medical trainees doing computational research, with literature answers grounded in verbatim, cited quotations. AI for Health (area 4; Aim 3): intraoperative imaging decision support, an advisory level label per C-arm image that the surgeon confirms and the stack withholds when its checks disagree.
Principal Investigator	Zhizhong Han, PhD, Assistant Professor, Department of Computer Science, College of Engineering, Wayne State University; founding director of the Machine Perception Lab; AIDaS affiliated member. Contact: [DATA NEEDED: office, phone and e-mail].
Co-Principal Investigator	Loren Schwiebert, PhD, Professor, Department of Computer Science, College of Engineering, Wayne State University; AIDaS affiliated member. Contact: [DATA NEEDED: office, phone and e-mail].
Clinical collaborator	Gregory Schwing, MD, PhD, General Surgery, Wayne State University School of Medicine / Detroit Medical Center; leads the Open Spine Consortium. Receives no award funds.
New Investigator status	The PI is a New Investigator: an Assistant Professor at Wayne State since January 2021 [DATA NEEDED: confirm that the appointment is tenure-track], and an NIH Early Stage Investigator until November 2027, ten years after the PI's PhD of November 2017 [DATA NEEDED: confirm the end date in eRA Commons]. The Co-PI is a Professor and not a New Investigator.
Request	$18,568 for 12 months, January 1 to December 31, 2027. No award funds go to an external collaborator: lines B2, B5, C1 and D buy goods and services, and Aim 3 uses WSU's own C-arm. The PI leads no other AIDaS proposal this cycle [DATA NEEDED: confirm].

Abstract. Wrong-level spine surgery persists because surgeons confirm the level by counting vertebrae on an X-ray from a C-arm, the operating room's mobile fluoroscope. The count fails in patients with an extra or missing lumbar segment and on images that show too few vertebrae. The Open Spine Consortium at Wayne State has built the parts of an automated check: 802 CTs with every vertebra labeled, a renderer that turns them into simulated C-arm images, and a level model. Never trained on a C-arm image, it names the level of 96.4 % of the lateral vertebrae a film-trained detector finds on a public C-arm dataset (88.0 % counting those it misses). Detection is the gap this project attacks first: a second detector, trained on two other devices' frames, finds 80.5 % of the vertebrae on that dataset's C-arm (90.6 % at a lower confidence threshold), against the 97 % set in advance. Closing the gap takes hundreds of GPU experiments on a shared computing cluster, which a person now writes, submits and repairs by hand, and which the medical trainees who hold the clinical questions cannot write at all. Aim 1 builds and benchmarks an AI agent that writes, submits and repairs its own cluster jobs and accepts a result only when a pass/fail test fixed before the run passes. We expect it to finish at least 80 % of 30 unseen tasks without human help, against under half for the same model working in one shot. Aim 2 gives the agent to eight medical trainees and measures, task against matched task, what it adds over a coding assistant, the 2026 program's setup. Their literature questions go to a language model on the grid, which quotes Scopus abstracts verbatim, with citation and DOI, under Elsevier's text and data mining terms, and no passage reaches an outside AI service. Aim 3 turns the agent on the detection gap, with learned X-ray simulation, and tests the result on CT-paired cadaver images taken at WSU. For the clinical validation after the award, an IRB protocol awaiting submission would add about 4,000 Detroit Medical Center spine surgery patients' CTs, operative notes and radiology reports, with any transitional vertebra recorded. Deliverables: the agent and its benchmark (MIT license, with the tasks an invention-disclosure review clears), the cadaver set, an FDA Pre-Submission, and NSF CAREER and NIH R21 proposals.

Keywords. agentic systems; code generation; high-performance computing; medical education; fluoroscopy; 3-D Gaussian splatting; domain transfer; spine surgery

Project narrative

1. Significance and innovation

Surgeons confirm the level of a spine operation by counting vertebrae on an X-ray from a C-arm, the operating room's mobile fluoroscope. The count fails often enough to matter: a 27-hospital Japanese registry recorded wrong-level surgery in 0.12 % of 34,544 patients having degenerative spine surgery (0.42 % in the thoracic spine) [1], and about half of responding US neurosurgeons report at least one wrong-level operation [2,3]. It fails most on a transitional vertebra, part lumbar and part sacral: 12.9 % of 550 surgical patients had one, and their levels were miscounted in clinic in 54 % of them [4]. The best-studied aid, LevelCheck, registers the patient's preoperative CT to each image [5] in 17 to 72 s in an operating room [6], inside or at the edge of the 20 s to one minute surgeons said they would wait [7].

The Open Spine Consortium at Wayne State has built the parts of a check that needs no CT, only the C-arm image: CTSpinoPelvic1K, an open dataset of 802 CTs with every vertebra and the sacrum labeled [8]; a physics renderer that turns them into C-arm images; and a level model (Figure 3a). With no C-arm image in training, it names the level of 96.4 % of the lateral vertebrae a film-trained detector finds on a public C-arm dataset (Figure 4a). Detection does not yet carry to a new C-arm: on one held out of its training, our C-arm detector finds 80.5 % of vertebrae (90.6 % at a lower threshold), short of the 97 % set in advance (Figure 4b). Closing that gap takes hundreds of GPU jobs written, refused and repaired on a shared cluster (one results figure took 250 fine-tunes). The consortium's 2026 trainee program hit the same wall: it relied on mentors expert in Python, imaging and cluster computing [9].

Our premise is that the bottleneck has moved from the model to the experiments, and that an agent can run them if it treats the cluster's refusals as errors to repair and accepts no result until a test fixed in advance passes. Three aims test it (Figure 1). **Aim 1** builds an agent that writes, submits and repairs its own jobs on the WSU grid. Its methodological contribution is a *gate-conditioned repair loop that treats the scheduler as part of the runtime*. Cluster refusals (a wrong quality-of-service, or QoS, setting; a job killed for memory) and code errors enter one loop that edits and resubmits the job. The loop stops not at a clean run but when the job's gate passes; a gate is a pass/fail test, sealed by hash before the first job, that the agent can read but not change. Agent benchmarks score code edits (SWE-bench [10]) or machine-learning engineering on a dedicated machine (MLE-bench [11], RE-Bench [12]). None we know of scores an agent against a shared scheduler's refusals and queue, under a fixed GPU budget, with gates fixed in advance. **Aim 2** puts the agent in medical trainees' hands through Beethoven, the consortium's browser extension (Figure 2). **Aim 3** turns it on the detection gap, with a paired CT and C-arm cadaver set imaged at WSU and a differentiable 3-D Gaussian X-ray renderer, like those of [13,14], built on the PI's Gaussian-splatting work.

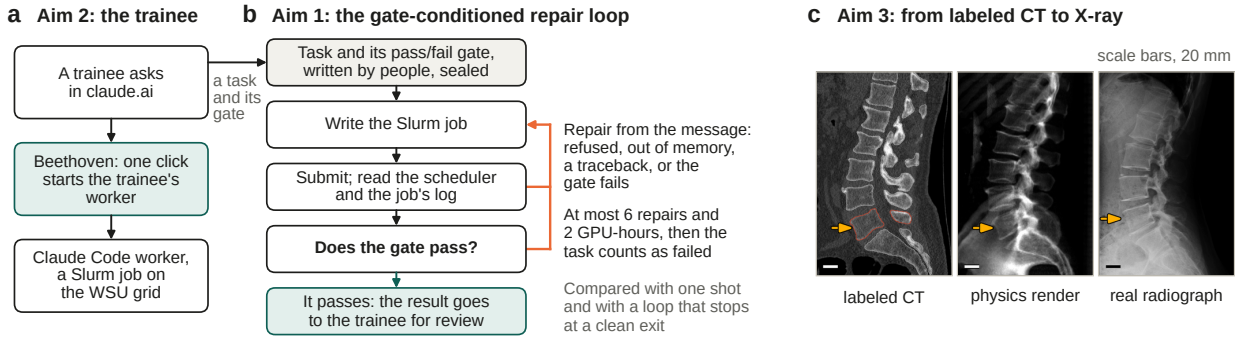


Figure 1. One loop, three aims. (a) Beethoven starts the trainee's agent. (b) The agent writes, submits and repairs the job until its gate passes or its budget runs out. (c) Why counting fails: a sixth lumbar vertebra (arrow) in a labeled CT, its render and another person's radiograph.

2. Aims, approach and preliminary work

Aim 1 (Foundational AI): a self-repairing agent for the grid. The agent is Claude Code, an AI coding agent, at a pinned model version, run by a model-agnostic test harness with no intervention (no human edit or resubmission). *Approach.* The benchmark holds 40 tasks (10 for development, 30 sealed until the final run) from Aim 3's experiments and the consortium's task lists (render banks, detector fine-tunes, figure rebuilds), each with a gate shown to pass on a reference solution and to fail on a planted error. A hidden check on each sealed task, unseen by the agent, catches gamed gates. Three arms share one model and one budget per task (six repairs, two GPU-hours, one queued job at a time): one shot, retried from scratch within the budget; a repair loop that stops at a clean exit; and the gated loop, which stops only when the gate passes. Each arm runs each task three times (360 runs). *Preliminary work.* The consortium's agents run through Beethoven as jobs under Slurm, the grid's scheduler; our prototype runs the loop on a simulated one with the grid's partitions and refusals, a design check, not a result. *H1:* the gated loop passes at least 80 % of the 30 sealed tasks with no intervention, no rule violation and no failed hidden check, against under half in one shot, and more than fresh retries or the repair loop, whose clean exits may fail their gate (false successes).

Aim 2 (Education and Workforce): medical trainees drive the agent. *Approach.* Eight medical students and residents in two waves of four [DATA NEEDED: confirm] each receive a grid account, Beethoven and a gated sub-project, their own or from Aim 3's list. The agent writes and repairs the jobs; the trainee owns the question and the result; a mentor reviews the code. Trainees' literature questions go to a language model on the grid that quotes verbatim, with citation and DOI, from an index of spine-related Scopus abstracts, under the text and data mining and AI rights Elsevier approved for WSU (full text awaits a ScienceDirect amendment); no passage trains a model or reaches an outside AI service, Claude included; a misquoted passage fails a gate, and the mentor marks any quotation that does not support the answer. In week 1 each trainee does two matched starter tasks outside the agent's development and benchmark: one with the agent, one with the 2026 setup (Claude as a coding assistant, the trainee submitting the jobs), task and order counterbalanced. The 2026 comfort survey is given before and after each wave; an IRB determination is requested in month 1, as in 2026. *Preliminary work* [9]: in 2026, 16 students, seven with no coding and eleven with no imaging experience, worked on ten sub-projects in that setup, with mentors expert in Slurm. All six domains improved on retrospective before-and-after ratings (Figure 2c), and all ten sub-projects went to a national meeting as abstracts. The report names what it lacked: a control group and objective measures. Aim 2 adds both. *H2:* with the agent, trainees pass more starter gates without a mentor's edit and reach each task's first passing GPU job sooner; with eight trainees an estimate with 95 % intervals that sizes the NSF CyberTraining proposal.

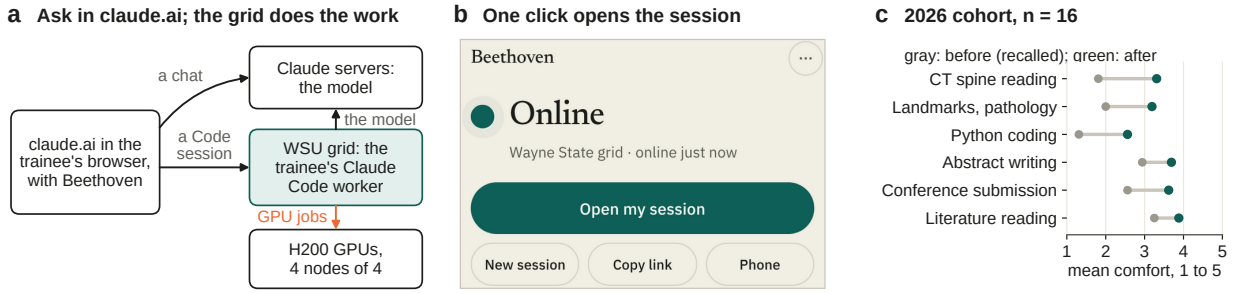


Figure 2. Beethoven, the trainee's door to the grid. (a) The trainee asks in claude.ai; the Claude Code worker runs as a Slurm job on the WSU grid and sends GPU jobs to H200s. (b) One click opens the session. (c) The 2026 cohort's mean comfort, before (recalled) and after (n = 16) [9].

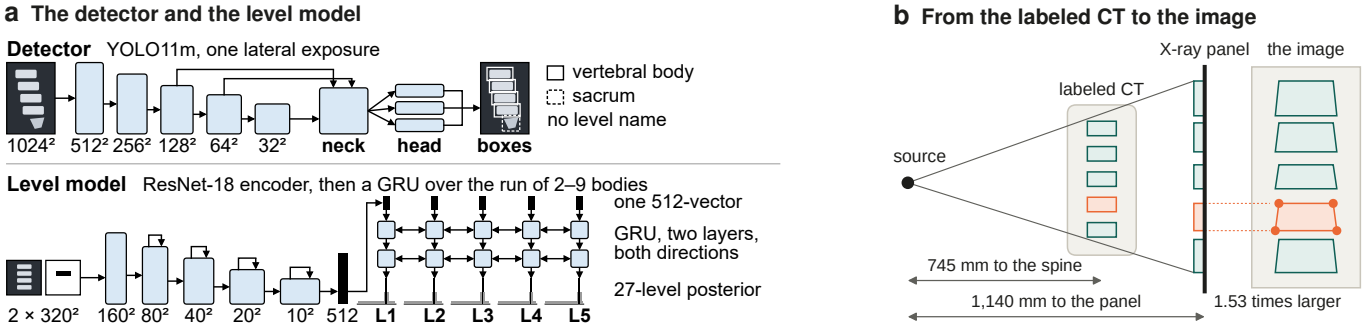


Figure 3. The level model, and how a CT labels the image. (a) The C-arm detector boxes the vertebral bodies and the sacrum, and the level model names each body among 27 levels. (b) Each labeled body and its corners project to the image; a transitional one in orange.

Aim 3 (AI for Health): labeled 3-D structure into 2-D fluoroscopy, at low latency. *Data.* Physics renders of CTSpinoPelvic1K, VerSe and TotalSegmentator CTs (7,045 standard and 1,482 transitional annotated frames) and a device-styled corpus of 142 TotalSegmentator spines; real frames of three devices: C-ArmLSpine (149 public lumbar cases, each a cone-beam CT with AP (front-to-back) and lateral frames, truth automatic: two CT labelers agreeing), DeepFluoro (cadaver pelvis fluoroscopy) and COBRA (Elekta radiotherapy imaging). *Model* (Figure 3a): a detector finds each body and the sacrum, a level model names each body, a count up from the sacrum checks the names, and the stack abstains where count and shape disagree or the name is unsure. New here, the C-arm detector will learn each body's four corners, as the film detector does, from labels made by projecting the CT's corners at the device's geometry (Figure 3b), exact on renders. *Preliminary results* (Figure 4). With no C-arm image in training, the level model names 96.4 % of the 617 lateral bodies the film-trained detector finds on C-ArmLSpine (88.0 % of all 676 with agreed truth) but 43.7 % of the 631 AP bodies. Fine-tuned on the AP frames of five cases, it names 90.8 %, and 98.1 % with all other cases, each test case held out. On the 233 frames where the sacrum is found (81.5 % of the 286 that show one), cross-validated models, checked against the count, name 98.2 % of 1,009 bodies, and the stack abstains on 9.9 % of those frames, catching five of the six it would have named wrongly. Detection does not carry: trained on two other devices' real frames, the C-arm detector finds 80.5 % of C-ArmLSpine's vertebrae and 76.8 % of its sacra at 0.74, a cross-device threshold chosen on other devices (90.6 % and 83.9 % at 0.25); on truth rebuilt after that scoring it finds 82.2 % of the 287 and 93.9 % of the 214 whose truth is complete and at least 80 % visible. *Closing the gap.*

Today's renders hurt: added to real frames, styled renders lower detection from 80.5 to 63.1 % at 0.74 (90.6 to 87.8 % at 0.25), and a classifier tells them from real frames 99 to 100 % of the time. Physics simulation (DeepDRR [15]) with strong domain randomization has matched real-data training for hip landmarks in cadaver X-rays (SyntheX [16]). We randomize devices' geometry and spectra likewise and hold out each device in turn. We also fit the Gaussian renderer to a subset of the labeled CTs, learning only spectrum, scatter and sensor response until that classifier is at chance, so the anatomy and its labels stay exact. The agent runs every Aim 3 job; GPU-hours and human hours per passed gate are Aim 1's field measures. *Cadaver set.* Two specimens from WSU's Body Bequest Program are CT-scanned and imaged AP and lateral at lumbar and thoracic positions on WSU's C-arm. A frame grabber records the video output, the set is sealed by hash before any model sees it, and registering each specimen's CT to its frames [5] projects the CT's labels onto every frame. The intended use is an advisory level label on each C-arm image, which the surgeon confirms. End-to-end latency is unmeasured, and Aim 3 measures it first. *H3:* on each held-out C-arm the detector finds 97 % of vertebrae and of sacra with no more false detections than today's at 0.74; the stack names a frame in under 1 s on a workstation GPU; and on every cadaver frame it names the right level or abstains.

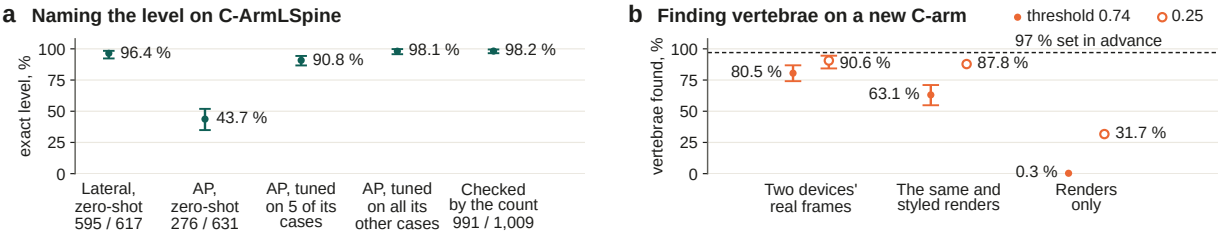


Figure 4. Held-out results; bars are 95 % case-clustered intervals, where recorded. (a) Level named exactly: zero-shot points never saw C-ArmLSpine, the others hold out whole cases. (b) Vertebrae found, of 287, on a C-arm never trained on, at 0.74 (filled) and 0.25 (open).

3. Evaluation: data, metrics and design

Aim	Data	Metrics	Design and analysis
1	40 tasks from the consortium's lists and Aim 3 (10 for development, 30 sealed), each gate checked on a reference solution and a planted error	pass rate by task type; repairs; interventions; rule violations (a job run on the shared login node, a credential in a file); GPU-hours per pass; false successes; gamed gates	three arms, one model, one budget per task; three repeats; paired bootstrap over tasks
2	8 trainees in two waves [DATA NEEDED: confirm]; two matched starter tasks each	starter gates passed without a mentor's edit; hours to each task's first passing GPU job; misquoted or unsupported quotations; comfort (secondary)	within-trainee crossover, task and order counterbalanced; plan preregistered
3	C-ArmLSpine, DeepFluoro and COBRA frames; the render banks; the sealed cadaver set	vertebra and sacrum recall at 0.74 and 0.25, with false detections; exact level; abstention; corner error (mm); latency per exposure (median, 95th percentile); render-to-real separability	leave-one-device-out, intervals clustered by case; the cadaver set opened once, after code and weights are hashed

H1 to H3 are written before any run; none is a powered test. H1 and H2 size the CAREER and CyberTraining proposals.

4. Expected outcomes, deliverables and timeline

Within the award: the agent, benchmark and three-arm report; the cohort's outcomes; the cadaver set, deposited beside CTSpinoPelvic1K as the Body Bequest Program's terms allow, and a transfer and latency manuscript; an FDA Pre-Submission; and the CAREER and R21. The R21's clinical data come from the consortium's pending protocol IRB-26-03-8648: about 4,000 DMC spine surgery patients over ten years, de-identified CT paired with operative notes and radiology reports, built outside the award. The seed buys what does not exist yet: the researcher's hours, the agents' subscriptions, the cadaver set and regulatory review. The trainee program, grid, pipelines and faculty time come in kind; the Gaussian renderer is fitted only as a feasibility test, its full build left to the R21. *Quarter 1:* agent version 1, gates sealed, IRB determination, recruitment, bookings. *Quarter 2:* benchmark run, wave 1 (months 4–7), cadaver imaging, transfer and latency results, Pre-Submission. *Quarter 3:* CAREER, wave 2 (months 8–11), the renderer's fit. *Quarter 4:* R21, final report, symposium. *Risks.* If the agent is not ready by month 4, wave 1 starts on the 2026 setup and the crossover moves to wave 2; a delayed specimen leaves the held-out devices as the test; a transfer result under 97 % becomes the R21's first aim; budgets cap each task's cost.

References. [1] Furuya M, et al. Bone Joint J 2026;108-B(5):673-80. [2] Mody MG, et al. Spine 2008;33(2):194-8. [3] Groff MW, et al. World Neurosurg 2013;79(3-4):585-92. [4] Nakagawa T, et al. Tohoku J Exp Med 2017;242(3):223-8. [5] Otake Y, et al. Phys Med Biol 2012;57(17):5485-508. [6] Manbachi A, et al. Ann Biomed Eng 2018;46(10):1548-57. [7] De Silva T, et al. Spine 2016;41(20):E1249-56. [8] Open Spine Consortium. CTSpinoPelvic1K, release 11. Zenodo 2026, doi:10.5281/zenodo.22797014. [9] OpenSpineConsortium: an open-source framework for medical student engagement in computational spine imaging research. Cureus 2026;18(7):e112661. [10] Jimenez CE, et al. SWE-bench. ICLR 2024. [11] Chan JS, et al. MLE-bench. ICLR 2025. [12] Wijk H, et al. RE-Bench. ICML 2025; arXiv:2411.15114. [13] Cai Y, et al. X-Gaussian. ECCV 2024. [14] Zha R, et al. R^2-Gaussian. NeurIPS 2024. [15] Unberath M, et al. DeepDRR. MICCAI 2018. [16] Gao C, et al. Nat Mach Intell 2023;5:294-308.

AIDaS funding priority alignment

Target agencies, programs and solicitations

1. **NSF CAREER** (NSF 22-586; CISE, Division of Information and Intelligent Systems), PI Han, built on Aims 1 and 2: the gate-conditioned agent as the research plan, the trainees as the education plan. Aim 1's tasks include the 3-D-to-2-D experiments the PI leads in Aim 3, so the agent serves the PI's own vision program. Deadline July 28, 2027; eligibility is judged at the deadline (an untenured Assistant Professor with no earlier CAREER award) [DATA NEEDED: the PI's tenure timing and earlier CAREER competitions]. If the PI is not eligible, the same plan goes to IIS's core research under CISE's Future CoRe solicitation (NSF 25-543).
2. **NIH R21 at NIBIB**, built on Aim 3: the Trailblazer Award for New and Early Stage Investigators if NIBIB reissues it (forecast FOR-EB-26-001), or else NIBIB's Exploratory/Developmental Bioengineering Research Grant (PAR-25-321) or the parent R21 (PA-25-304), both open through October 2027. Submitted October 18, 2027 (October 16 is a Saturday), while the PI is still an Early Stage Investigator. Preliminary data: the cadaver set, the held-out-device result and the measured latency; its clinical validation, shaped by FDA's Pre-Submission feedback, draws on the consortium's pending protocol IRB-26-03-8648 (about 4,000 adult DMC spine surgery patients, their CT and records), where 99 patients give 80 % power to show a Wilson lower bound of 90 % if the true accuracy is 97 %.
3. **NSF CyberTraining** (NSF 23-520), a Pilot of up to $300,000 over two years led by Co-PI Schwiebert: agent-assisted training on advanced computing, sized by Aim 2's estimates. Deadline January 20, 2028; submitted in any case.

Relevance to national science and technology priorities

America's AI Action Plan (July 2025) asks for investment in AI-enabled science and, in its worker-first agenda, for AI skills in the workforce, which Aim 2 builds in Michigan's medical trainees. Executive Order 14363 (November 2025) launches the Genesis Mission, which joins AI agents to national computing, and NSF's Dear Colleague Letter 26-023 invites proposals to existing programs that support it. NIH's Bio Genesis Mission carries it into biomedicine. Aim 1 is that agenda at a university's scale: an agent that runs science on shared computing and is held to verifiable gates. Executive Order 14303, Restoring Gold Standard Science (May 2025), and OSTP's guidance under it (June 2025) ask agencies to promote pre-registration of study protocols and transparent reporting of null results; every gate in this project is pre-registered by hash, and failed gates are reported. NIH's data sharing policy and its Strategic Plan for Data Science 2025–2030 ask for data others can reuse, as CTSpinoPelvic1K is and the cadaver set will be where its terms allow. FDA's Q-Submission program gives Aim 3 written feedback on its validation plan within the award.

Team, expertise and roles

The PI directs the project, writes the FDA Pre-Submission, leads the CAREER and R21 proposals, and builds the agent and the 3-D-to-2-D model, including the Gaussian X-ray renderer; a computer vision researcher, the PI leads the Machine Perception Lab and publishes on 3-D Gaussian splatting and neural implicit surfaces (CVPR 2025 and 2026, NeurIPS 2024). The PI's part in the consortium's work so far: [DATA NEEDED: the PI's role, and since when]. The Co-PI, a high-performance computing researcher who accelerates scientific codes on GPUs, builds the benchmark harness and grid integration; the Co-PI and the clinical collaborator have published four high-performance computing papers together (2021–2024). The clinical collaborator leads the Open Spine Consortium, whose dataset, models and 2026 trainee program [9] this proposal builds on, and runs the cohort, the IRB determination, the imaging session, the literature index and the pending protocol. Line A's part-time researcher runs the benchmark, logs the measures and curates the cadaver set. Effort: PI [DATA NEEDED: %], Co-PI [DATA NEEDED: %], in kind.

Timeline for external submission

When	Milestone
January–March 2027 (months 1–3)	One-page concepts to the NSF IIS and NIBIB program officers; benchmark and gates sealed; the Pre-Submission package written and reviewed
April 2027 (month 4)	FDA Pre-Submission filed; written feedback due within 70 days
July 28, 2027 (month 7)	NSF CAREER submitted, with Aim 1's benchmark and Aim 2's first wave
September 2027 (month 9)	The sealed cadaver set opened, once, for the R21
October 18, 2027 (month 10)	NIH R21 submitted, with the cadaver set, the latency and FDA's feedback
December 2027 (month 12)	Project ends; AIDaS final report and symposium
January 20, 2028 (month 13)	NSF CyberTraining Pilot submitted, Co-PI leading
2028	R21 resubmission at the first standard date after its summary statement; CAREER resubmission July 26, 2028, if eligible; NSF Smart Health when its next solicitation opens

The CAREER and R21 meet the expectation of a federal proposal within 12 months of the project's end, and the CyberTraining Pilot follows whatever their outcome; publications and proposals from the award acknowledge AIDaS.

Budget and justification

Line	Item	Basis	Amount
A	Part-time researcher, hourly	520 h at $15.00: Aim 1 180 h, Aim 2 160 h, Aim 3 180 h	$7,800
A2	Fringe on line A	FICA, 7.65 %	$597
B1	Two cadaver specimens, Body Bequest Program	placeholder $1,250 each [DATA NEEDED: program fee]	$2,500
B2	Specimen CT, two scans	placeholder $300 each [DATA NEEDED: facility and rate]	$600
B3	C-arm session, 0312 Elliman, through DLAR	4 h at a placeholder $300 [DATA NEEDED: DLAR rate; that the room takes human specimens]	$1,200
B4	Dosimetry and OEHS review	placeholder [DATA NEEDED: OEHS fee]	$300
B5	Frame grabber for the C-arm's video output	Magewell USB Capture AIO, retail price	$619
C1	Claude Team Premium seats for the agents	6 seats billed annually at $180 each	$1,080
C2	Data management and storage	2 TB of C&IT storage at $400 per TB-year; Zenodo free	$800
D	FDA Pre-Submission: regulatory consultant	10 h at $300; FDA charges no fee	$3,000
E	Anthropic AI for Science application	no fee; written by the PI, in kind	$0
F	AIDaS symposium poster	36 by 48 in at $6 per sq ft	$72
	Total requested		**$18,568**

- **A, A2.** An hourly graduate student assistant (the call's graduate student support), or else a post-baccalaureate researcher: ten hours a week for 52 weeks at $15.00, Michigan's minimum wage from January 1, 2027 (MCL 408.934) [DATA NEEDED: WSU's job category and scale]. Temporary employees and students are not benefit-eligible, so A2 is FICA alone: 7.65 % for a hire who is not an enrolled student; hiring a student, at WSU's composite rate of 1.7 % (fiscal year 2025), would buy about 30 more hours. Aim 1's hours are 360 runs at half an hour; Aim 2's go to the cohort's accounts and measures; Aim 3's are 100 curating the cadaver set (de-identifying, labeling, sealing by hash) and 80 checking the agent's leave-one-device-out and latency runs against their gates. At $18 an hour the total would be $20,247, over the cap unless the hours fall to 507.
- **B1 to B5.** No public dataset we know of pairs a diagnostic CT of the thoracolumbar spine with C-arm frames of the same body: C-ArmLSpine pairs lumbar cone-beam CTs with frames posed by edge registration, and DeepFluoro pairs cadaver CTs with fluoroscopy of the pelvis. The cadaver set adds WSU's own C-arm. The Body Bequest Program's donor authorization permits de-identified images [DATA NEEDED: the program's terms for public release]. Each C-arm user enrolls in WSU's dosimetry program and files the training form of OEHS, WSU's safety office, both with the director of DLAR 15 business days before the session. The frame grabber records what the C-arm shows, the input a deployed tool would read [DATA NEEDED: the C-arm's model and video output]; at $619 it is far under WSU's $5,000 capital threshold and becomes WSU property.
- **C1, C2.** Anthropic's Team plan for scientists lets a verified PI add Premium seats at $15 a month, $180 a year billed annually: four for the trainees, passed from wave 1 to wave 2, and two for the benchmark and development. All six are bought in January 2027, since the price holds 12 months from sign-up [DATA NEEDED: P-card route and the PI's verification]. The WSU grid's compute is free to WSU researchers and given in kind. Storage holds the render banks, the cadaver set and every benchmark run's logs for the award year; the data go to Zenodo at its end.
- **D.** FDA charges no fee for a Pre-Submission filed before October 1, 2027, when a fee proposed for fiscal year 2028 would begin. The filing asks FDA whether an advisory level label on each C-arm image, which the surgeon confirms, is the right intended use; whether the held-out-device and cadaver tests suffice as analytical validation; and what clinical study, against what reference standard, the R21 should plan. A mid-level consultant reviews the PI's package in months 1 to 3.
- **E, F.** Anthropic's AI for Science program takes applications for up to $50,000 in API credits per project and charges nothing; credits would add benchmark repeats, on tasks its data terms allow. Presenting at an AIDaS symposium is an award expectation; the poster is printed at the School of Medicine's Medical Communications.
- **Not requested.** PI and Co-PI salary, travel, tuition, capital equipment and indirect costs [DATA NEEDED: confirm AIDaS applies no F&A].
- **Allowability and dependencies.** The call's allowable costs include graduate student support, computing and data storage: lines A and A2 (with a graduate student assistant), C1 and C2, $10,277 in all, above the call's smallest award of $10,000. Lines B, D and F are research costs outside those headings [DATA NEEDED: the AIDaS office's answer]; B2 to B5 depend on B1. Without B the request is $13,349: Aim 3 tests on the held-out devices' frames, and its 100 cadaver hours go to benchmark repeats; without B and D, $10,349; without B, D and F, $10,277.

Biosketch: the Principal Investigator

The two-page NSF biosketch of the Principal Investigator goes here in the submitted PDF. Put the file below beside `main.tex` and compile again, and it replaces this page.

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Biosketch: the Co-Principal Investigator

The two-page NSF biosketch of the Co-Principal Investigator goes here in the submitted PDF. Put the file below beside `main.tex` and compile again, and it replaces this page.

`biosketch_schwiebert.pdf`